

AKADEMIKA

**PCB-B
BASIC PCB LAB**

**EXPERIMENTAL & SERVICE
MANUAL**

.... A MESSAGE FROM

It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO;

- Light years ahead – refers to leadership.

As leaders in our industry in India, we are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that we have complete, truly innovative & superior training products. We are so committed to quality that we back our products with a complete comprehensive warranty.

AKADEMIKA LAB SOLUTIONS

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SAFETY RULES

Carefully follow the instructions contained in this handbook as they give important points on safety during the installation, use and maintenance. Keep this handbook always with you for any further help.

After the unpacking, set all accessories in order so that they are not lost and check the kit integrity. In particular, check that the kit is integral and shows no visible damage.

Before connecting the power supply to the kit, be sure that the jumpers and patch chords are connected correctly as per experiment.

This kit must be employed only for the use for which it has been conceived, i.e. As educational equipment and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of any fault/bad operation, turn off the power supply and do not tamper the kit. In case servicing is required, contact the service center for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

ETCHING TANK

Use plastic tongs to place PCB into tank. Do not use hands / fingers as the chemicals are dangerous.

Wear goggles/visor and appropriate safety clothing when placing PCB in the tank and also when removing it. Chemicals are dangerous if 'splashed' on to the user.

PCB DRILL

Ensure that the PCB is clamped / held securely so that the drill does not 'spin' it round causing an accident to hands / fingers.

Wear goggles when drilling in case the PCB shatters or pieces are thrown off into the face.

WARRANTY

This PCB LAB is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like chemical lab etc.

This warranty is subject to the following conditions and limitations. The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on Akademika Lab Solutions's sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the warranty period if the service is needed, purchaser should get in touch with the service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. Of kits has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika Lab Solutions's judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, AKADEMIKA. Bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika Lab Solutions Does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions. Is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or sales outlet.

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TECHNICAL SPECIFICATIONS

PCB-B: BASIC PCB LAB

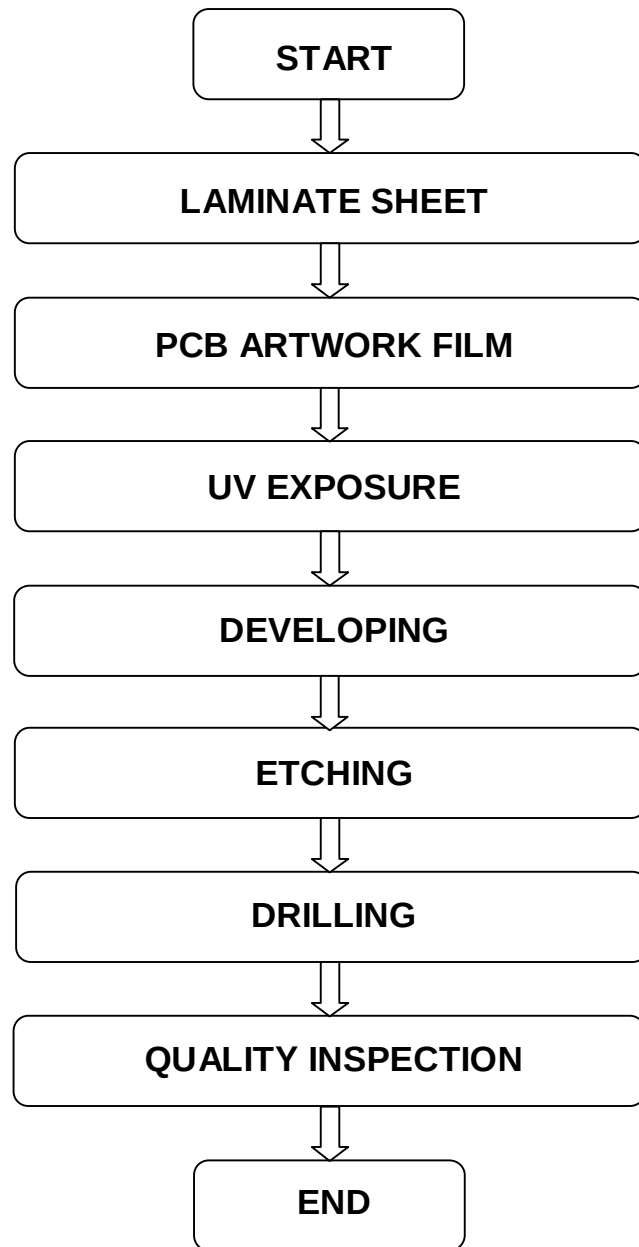
UV Exposure Box	:	Supply Voltage.-: 240V, 50Hz Power (Max.) -: 200W UV Source:-: 6X15W Fluorescent Lamp Mask Area -: 22 X 37 cm
Etching tank with pump & heater	:	300W / 230Vac Spare heater Tank Capacity -: 2L Dimension -: 360x280x40mm
Drilling Machine	:	230Vac

BASIC PCB LAB EXPT& SER

Manual : 01 No.

Power Supply : 01 No

PCB Manufacturing Process Flow Chart



INTRODUCTION

FEATURES:

PCB is the abbreviation for Printed Circuit Board. Some may refer it as PWB which stands for Printed Wiring Board. The board consists of copper tracks connection printed on the surface of a non-conducting material. Your designed circuit connection will be contained within this board. This method of fabrication is an efficient method for reproducing a circuit. The circuit quality can be more consistent and the reproduction can be faster. This makes it suitable for manufacturing large quantities of electronic circuits.

PCB allows you to produce custom prototype quality circuit boards in a few hours with inexpensive equipment. Short development cycles have previously been restricted to the domain of software projects. Hardware designers can now benefit from improved, if still somewhat slow, cycle times.

Introduction to BASIC PCB LAB

UV EXPOSURE BOX



The UV exposure box has a maximum exposure area 230 x 230mm. It includes digital timer & vacuum table.

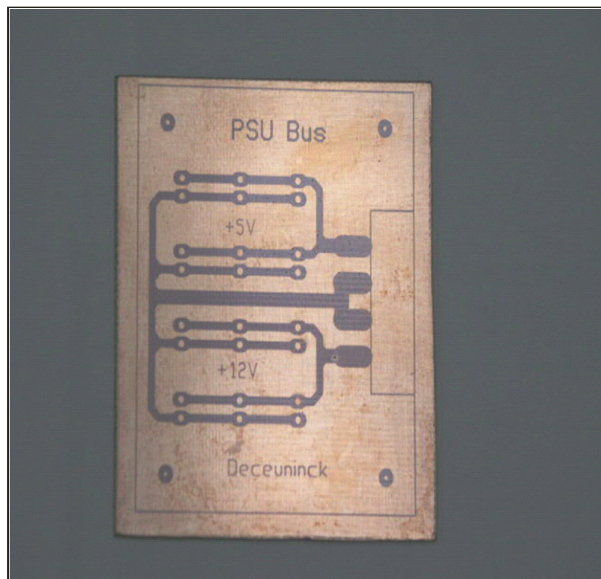
The peak wavelength of the UV lamps is 350~400nm.

Etching tank



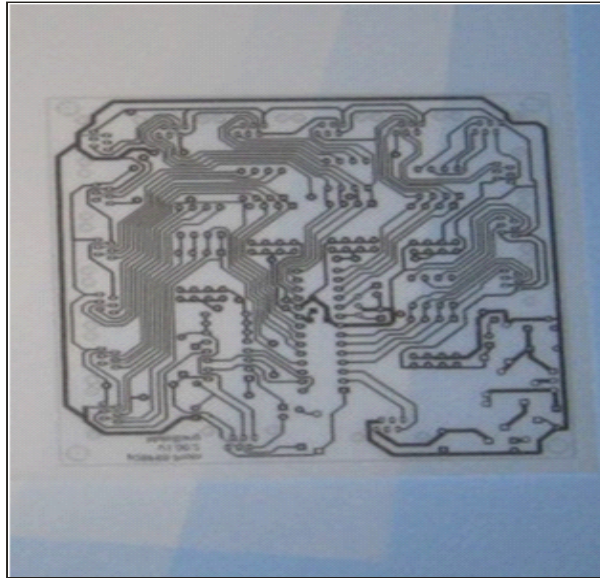
Etching tank is a thin vertical format to allow minimum chemical usage. include a thermometer and clip to suspend the PCB in the liquid etchant includes air pump and heater. The air pump allows the etchant to be bubble agitated, and the heater brings the etchant to the optimum temperature, speeding the etching process. It is recommended that the pump be supported at a position higher than the liquid level to prevent siphoning if the pump is switched off.

PCB Development



If the resist does not fully remove from the exposed areas, the UV received was insufficient. Possible causes are: insufficient exposure time; the UV source is old and not producing enough UV; incorrect UV frequency (use 350~450nm UV source); the artwork is not UV transmissive.

PCB ARTWORK



When producing artwork for making PCB's, some materials produce better results than others.

Overhead transparency sheets usually produce very poor results, as they cannot aggregate the ink or toner density enough to produce high contrast. Some materials are not UV transparent, and are also unsuitable for using as PCB artwork.

Using a suitable base media as provided with an inkjet or laser printer, good results can be obtained.

PAPER

Paper artwork base is not recommended for inkjet printers, and the resolution is lower than polyester. Paper artwork base minimum resolution is 0.15mm lines with a 0.25mm gap between lines, subject to printer capability. It is most suitable for single use, with a laser printer. It is currently available in A4 size supplied in bags of 10 sheets.

PCB LAMINATE SHEET

LOW COST, EASY TO WORK

The positive resist coated printed circuit board (PCB) are used for many prototyping, educational and low volume applications.

Phenolic substrate PCB is particularly suitable for less demanding applications. We only supply it with 1oz copper cladding on one side. The phenolic substrate is colored brown, and is commonly seen in older electrical equipment. It is easier to cut and drill compared to fiber-glass PCB

However, it is not as robust, and is not suitable for use in situations where vibration or flexing is present. Care in soldering is recommended, as excessive soldering temperature or times can lead to charring more readily than fiber-glass substrate.

The positive resist has good exposure latitude and can produce very detailed track layouts when a suitable artwork and processing is used.

ETCHANT

We recommend the supplied etchant for use with our boards. If using an three EB-750 bags are required for to the tank

Mix the bags with warm water until the powder is dissolved. Pour the solution into the etching tank, ensuring that the level is above the heater element. Using the air pump to circulate the solution, turn on the heater until the solution is at the operating temperature. As the etching process increases the etchant temperature, larger PCBs (>300cm² etching area) the temperature should be set to 40~50°C, and smaller PCB's to 50~55°C. The greater the amount of etching, the higher the temperature rise. The maximum etchant temperature should be limited to 60°C.

Lower the boards into the liquid for etching. Carefully remove the boards and rinse

EXPERIMENT NO.1

EXPERIMENT: 1

NAME

PCB EXPOSURE

OBJECTIVE

TO STUDY FOR PCB EXPOSURE PROCESS

THEORY

The UV exposure box has a maximum exposure area 230 x 380mm and can be used for single sided PCB's.

It includes digital timer & vacuum table.

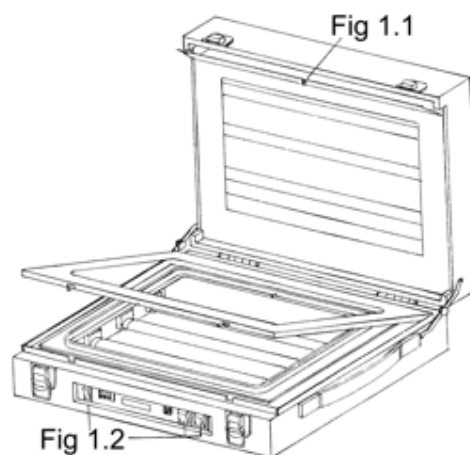
Three tubes top and bottom ensure high intensity ultra-violet radiation for rapid exposure time.

EQUIPMENTS

- 1) UV Exposure Box
- 2) PCB Laminate
- 3) PCB Artwork Positive Film

PROCEDURE

The power cable is located in the top lid of the exposure box, to release pull small brass knob. (Fig 1.1)

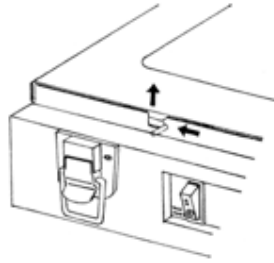


Turn off all switches on the front panel (fig 1.2) & plug power cable. Ensure the voltage supply is correct and plug into electricity socket.

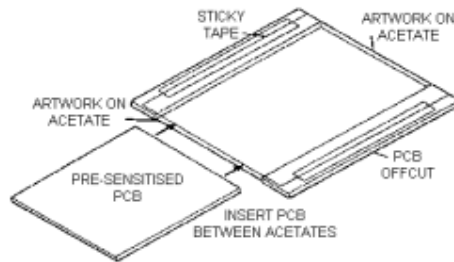
Adjust exposure time using **second preset** buttons.

Typical exposure time is 60-90 seconds for photo plot and 70-120 seconds for artwork base when used with the supplied PCB.

Unlock the vacuum clamp by pushing the metal latches outward with your thumbs, and lift clamp.

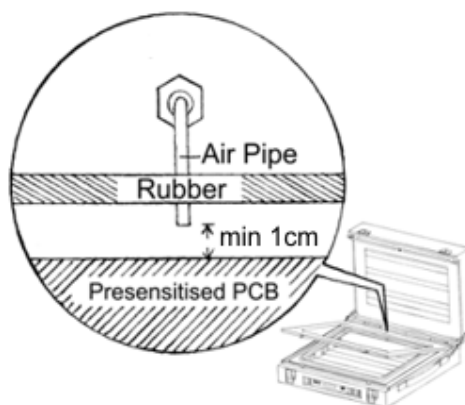


Remove protective film from presensitised PCB and align the artwork with the PCB. Alignment of double sided PCBs is made easier by using PCB off cuts to make a pouch with the artworks pre-aligned as shown in the image.



Artwork should be printed as a positive (black tracks) reversed image. Lay the printed side against the PCB, so the result will be correct.

Place the assembled PCB and artwork inside the vacuum clamp, leaving a minimum 1cm distance from the air pipe.



Close and latch the vacuum clamp. Start the vacuum pump and wait for approximately 30 seconds until the air under vacuum clamp has been fully removed.

Close box and turn on UV light source upper/lower or both as required. Display panel counts down seconds and at the end of exposure time, a buzzer will give a 2-3 seconds signal.

Turn off vacuum pump, open the box and wait a few seconds, then unlock the vacuum clamp and open it slowly. Remove and process the PCB. PCB is a printed circuit board

OBSERVATION

CAUTION:

1. Avoid UV exposure at a short distance for more than 5 seconds. If necessary, wear UV protective glasses.
2. Adjusting the second preset during exposure may interrupt the timer.
3. Unplug the power cable before replacing the fuse, 3 Amp for 100-120V, 2 Amp for 220-240V.
4. Do not use solvents on transparent plastic surface, use soft cloth with water/detergent or 50% water/alcohol.
5. Normal artificial room light will not harm presensitised surface if exposed less than 5 minutes.
6. Breaks and flaws in artwork may be touched up with a black text pen.

EXPERIMENT NO.2

EXPERIMENT: 2

NAME

PCB DEVELOPING

OBJECTIVE

TO STUDY FOR PCB DEVELOPING PROCESS

THEORY

A bag of DP-50 developer is mixed with 1 Litre of 25°C water. The exposed PCB is gently agitated in solution. The exposed areas of resist will dissolve into the solution, leaving the green resist in areas that were not exposed to UV light.

EQUIPMENTS

Exposure PCB Laminate, Plastic Tray, Chemical for Delivering, Water 1 Litre

PROCEDURE

A. Prepare the solution:

1. For one litre (1000 cc) of water into a small, flat plastic tray.
2. Put the whole pack of DP-50 powder (50g) into the water and slightly shake the tray until the powder is completely dissolved.

B. Development :

1. Place the exposed presensitized PCB in the solution and slightly shake the tray until the tracks area is clear and there is no more blue smoke like resin coming off the exposed area on the board.
2. Rinse the board under running water to remove the developer and to stop the developing process.
 - The correct ratio between developer and water is important but not critical, a maximum tolerance of -10% to +30% is allowed.
 - The normal developing time is between 30 seconds and 2 minutes. If the developing time is less than 30 seconds. The unexposed parts of the board will be easily damaged due to too concentrated solution.
 - The developing process should take place at room temperature. Higher temperature will shorten the developing time.
 - The usable time of solution is one day after the first time it is used.
 - The developing ability of each pack of DP-50 is about 20 PCBs (10 x 15 cm).

EXPERIMENT NO.3

EXPERIMENT: 3

NAME

PCB ETCHING

OBJECTIVE

TO STUDY FOR PCB ETCHING PROCESS

EQUIPMENTS

Developed PCB Laminate
Etching Tank with Pump and Heater
Water 1 Litre
TEMPRETURE METER

PROCEDURE

A. Preparing the etching solution:

1. Use etching tank
2. With etching tank add 3 packs of etching solution into the tank, add clear water to make up to 2250 c.c. (the 30 cm marking).
3. Stir with a glass, a wooden, or a plastic, till fully dissolved.

B. Etching:

1. Place the fully developed sensitizing board into the etching tank till it is completely etched, remove and wash clean with clear water (the whole processing entirely open to visual observation)

C. Important notes: guard against to high concentration of the etching solution:

1. Where the concentration of the etching solution goes too high, say prolonged disposition, evaporation, untoward proportioning, etc., blue crystal may develop on the bottom, and that might give an instance of dotted etching, therefore it is recommended that inspection may prior to each operation, should crystallization already takes place on the bottom, all that needs to be done is to supplement water to original level, and let the crystal remains on the bottom.

D. Additional notes:

1. Once pack of etching solution is good for etching 10-20 pcs of 10 x 15 single sided sensitizing board..
2. Solution gone for etching processing will turn blue.
3. The etching solution may produce bubble, but that is normal.
4. A higher etching temp. Will bring about a faster etching process, but in no case should the etching temp. Exceed 60⁰C. (The cooper foil while being etched will by itself release heat, thereby contributing to a rise in temp. At the same time)
5. As a rule it will take approx. 6 Mims to etch through using newly unpacked solution, with the temp. Of 50⁰C, where the etching is still incomplete beyond 45 minutes, it is necessary to replace a new pack.
6. The refuse etching solution must be disposed of with the blue refuse treatment agent offered separately, as to which a descriptive brochure is available and that deserves a referential reading.

EXPERIMENT NO.4

EXPERIMENT: 4

NAME

PCB DRILLING

OBJECTIVE

TO STUDY FOR PCB DRILLING PROCESS

EQUIPMENTS

Etching PCB laminate, Drilling machine, Carbide drill bits

THEORY

Always wear safety glasses when operating a drill press, especially if you are drilling with carbide PCB drill bits.

If available, always use a vacuum cleaner to remove debris and collect airborne dust during the drilling operation. The dust generated during PCB drilling can pose a very serious health hazard and should not be inhaled or ingested under any circumstances.

When using a conventional drill press, hole placement accuracy can be improved and drill breakage minimized through the use of a "sensitive drilling" or "finger" chuck. Small format, precision high-speed drill presses, ideal for PCB fabrication, are also available from a number of sources.

Regardless of the type of drill press being used, a pressure foot should be employed if available.

If available, position a work lamp on a flexible mount as close to the work surface as possible.

Although more brittle than conventional high speed steel (HSS) drills, tungsten carbide bits designed specifically for PCB drilling will yield far superior hole wall quality. Minimize burr formation, and outlast HSS bits almost 10 to 1. The downside is that, with smaller diameters [0.018"(0.46mm) and less], the carbide drills are easier to break and must be handled carefully.

Always use drill bits that have been fitted with depth setting rings. This will allow you to set the plunge depth stop on your drill press to a single value that will work for all bit diameters.

Prepare a chart that links the various diameter bits with the symbols used in the drillmaster.